Bouncing Ball Supplementary Material

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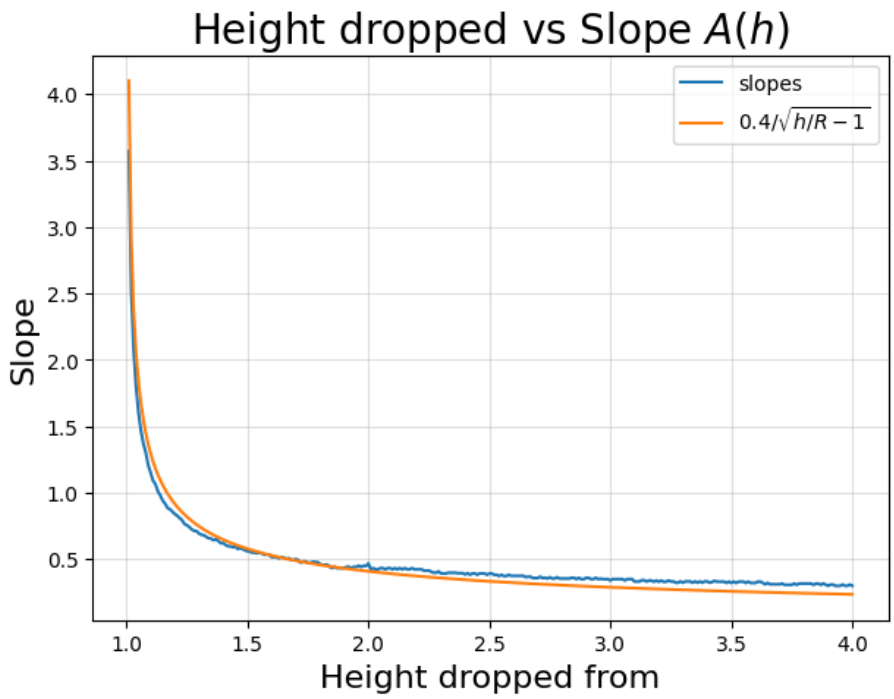

Figure 1: Dependence of the fitted slope $A(h)$ on the initial drop height h. The numerical values of $A(h)$ obtained from the simulations (blue) are compared with the analytical prediction $\frac{0.4}{\sqrt{h/R-1}}$ (orange).

The strong agreement between the numerical and analytical results supports the proposed inverse square-root dependence of $A(h)$ on drop height. However, more exploration is required to establish the exact analytical expression of this relationship.

Unlike the slope $A(h)$, the intercept $B(h)$ does not exhibit a clear systematic dependence on drop height and instead fluctuates. $B(h)$ may be weakly dependent on h, although further investigation is required.

Using these conclusions obtained from Figures 1 and 2, we can improve our previous formula for number of bounces to,

$$N(d,h) = \frac{2}{5\sqrt{h/R-1}} ln\left(\frac{R}{d}\right) + 0.459 \tag{1}$$

Using this formula, and rounding to the nearest integer, we get a near-perfect prediction for the number of bounces a ball dropped at position $[d, h]$ above a sphere of radius 1.

The angle remains small for the initial bounces before increasing rapidly at higher bounces. As expected, since the initial velocity of the ball is predominantly normal to the sphere's surface, before gaining more tangential velocity at later bounces

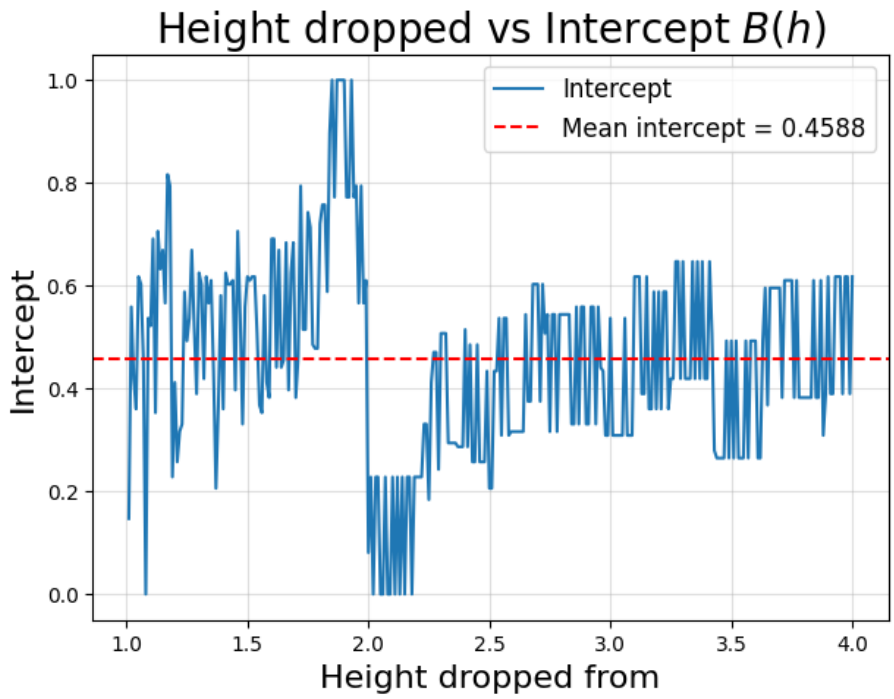

Figure 2: Dependence of the fitted intercept $B(h)$ on the initial drop height h. The numerical intercepts obtained from the simulations (blue) are shown together with their mean value (red dashed line).

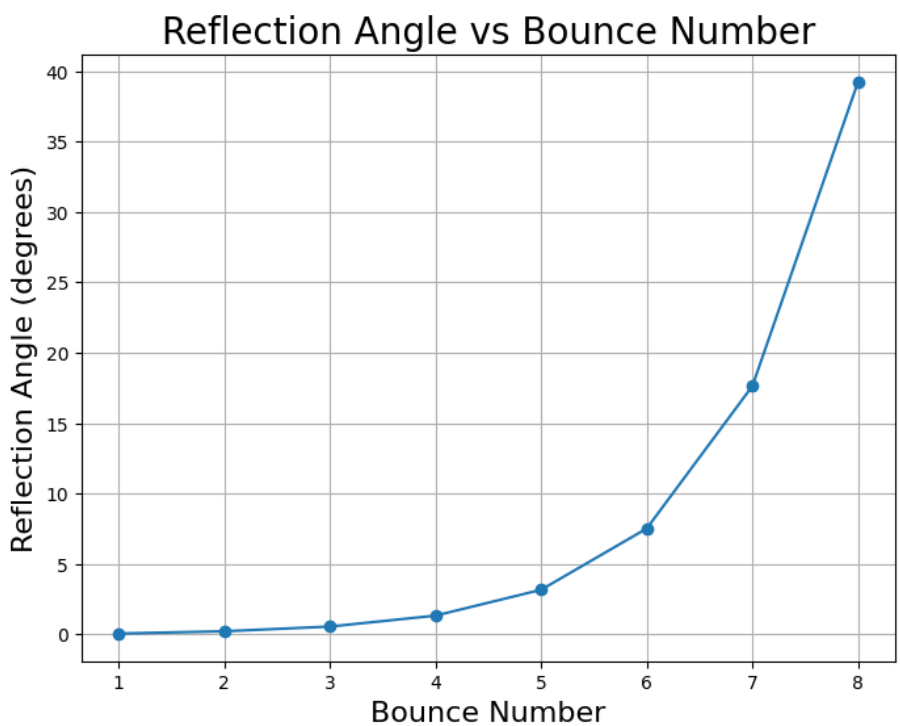

Figure 3: Reflection angle, relative to the surface normal at the point of impact, for each bounce on a sphere of radius 1. The ball is dropped from a height of $h = 1.1$ and with a horizontal displacement of $d = 0.001$. The coefficient of restitution is set to 1.